

MICROGRID SIMULATION WRITE-UP

System configuration:

- Generator: 60 kVA @ 0.8 PF → 48.0 kW prime
- PV array: 100 kWp (module parameters based on datasheet assumptions)
- Battery: 200 kWh, 60 kW power, KiBaM state update (current-based)
- Load: residential profile from Load_model.py (stochastic + seasonal)
- Timestep: 60 min (8760 steps)

Key performance indicators (annual):

- Renewable fraction: 47.8%
- Unmet load: 0.73%
- Diesel runtime: 5055.0 h
- Fuel consumption: 43356.5 L
- Operating cost: \$69527.05

Assumptions and modeling notes:

- * Generator fuel curve linearized (slope/intercept) from datasheet points
- * PV output uses Assignment 7 TAG/Kt model (solar time + stochastic variability)
- * Battery uses KiBaM kinetic model (current-based), with min SOC = 20%
- * Energy balance is maintained each timestep (error < 0.01 kW)
- * Dispatch strategy: renewables → battery (discharge/charge) → diesel → shedding

Output files generated:

- microgrid_results_*.csv (timestep dispatch and state)
- microgrid_metrics_*.csv (annual KPIs)
- microgrid_plot_*.png (dispatch plots)
- microgrid_writeup.pdf (this summary)